

# MLLM-Driven Automated Layer Labeling, Metrology, and Anomaly Detection in Semiconductor and Display Array Electron Micrographs

## Introduction to the Shifting Paradigm in Microelectronic Metrology

The relentless scaling of microelectronic architectures, ranging from advanced semiconductor logic nodes to complex Low-Temperature Polycrystalline Silicon (LTPS) display arrays, has driven structural geometries into extreme sub-micron and nanometer regimes. At these scales, the characterization of multi-layered thin-film stacks relies entirely on high-resolution imaging modalities, primarily scanning electron microscopy (SEM) and transmission electron microscopy (TEM).<sup>1</sup> These electron micrographs capture the physical reality of the fabricated device, revealing the precise morphological details of dielectric insulators, metallic interconnects, semiconductor channels, and organic planarization layers.<sup>2</sup> However, the extraction of actionable metrological data from these images has historically remained a profoundly manual, labor-intensive, and subjective process.<sup>1</sup>

In a conventional failure analysis (FA) or inline metrology workflow, a human engineer must meticulously scrutinize a micrograph, correlate the visual gray-level striations with a known schematic or design table, manually draw boundaries to annotate distinct layers, and utilize point-to-point software tools to estimate layer thicknesses.<sup>1</sup> Furthermore, the engineer must rely on domain expertise to identify morphological anomalies, voids, or delamination events.<sup>1</sup> This manual pipeline introduces substantial latency into the manufacturing feedback loop, with end-to-end failure analysis—from failure verification through physical analysis to root-cause reporting—traditionally consuming days to weeks of expert time.<sup>1</sup>

The emergence of Multimodal Large Language Models (MLLMs), which possess architectures capable of natively processing both complex textual inputs and high-resolution visual data, presents a transformative solution to this metrological bottleneck.<sup>6</sup> By leveraging models such as GPT-4 Turbo with Vision, LLaMA-3.2-11B-Vision-Instruct, and specialized domain-adapted frameworks like FabGPT and MicroscopyGPT, the industry is pivoting toward fully autonomous, agentic analytical workflows.<sup>6</sup> The overarching objective is to deploy these MLLMs to ingest a user-provided prompt—specifically, an image featuring a designated region of interest (ROI) enclosed in a red rectangle, accompanied by supporting textual or tabular data—and autonomously execute semantic layer labeling that mimics or exceeds human deductive reasoning.<sup>6</sup>

Beyond mere semantic labeling, the integration of these MLLMs with deterministic physical measurement algorithms, open-vocabulary segmentation foundation models, and multi-agent

orchestration frameworks facilitates the autonomous estimation of layer thickness and the execution of rigorous anomaly detection.<sup>1</sup> This comprehensive report delivers an exhaustive analysis of the cognitive, algorithmic, and computational mechanisms required to deploy MLLMs for automated electron micrograph analysis. By examining a specific LTPS array cross-section as a foundational case study, this analysis dissects the visual reasoning capabilities of MLLMs, the transition from traditional convolutional neural networks to advanced segmentation architectures, the mathematical basis of intensity profile metrology, and the deployment of agentic pipelines for prognostic failure analysis.

## Contextual Reasoning and Multimodal Data Fusion

The fundamental advantage of an MLLM over traditional machine vision systems lies in its capacity for generalized cognitive reasoning and its ability to synthesize disparate data modalities.<sup>8</sup> Traditional machine vision, relying on static pixel classifiers, fails when confronted with the ambiguity of electron micrographs, where different materials may yield identical gray-level contrast.<sup>11</sup> To execute automated layer labeling within a targeted ROI, the MLLM must replicate the holistic deductive process of a human engineer, which involves cross-referencing the visual image with structured supporting information.

### Analyzing the LTPS Array Case Study

To concretize the operational mechanics of MLLM-driven layer labeling, it is imperative to analyze a representative exemplar: a cross-sectional SEM image of a complex LTPS display array, featuring a user-defined red rectangle indicating the region to be labeled. The micrograph displays a highly stratified architecture comprising over fourteen distinct thin-film layers deposited on a glass substrate. Crucially, the image is accompanied by a supporting data table detailing the structural hierarchy, material composition, and nominal design thickness of each layer.

The structural hierarchy detailed in the supporting information, ascending from the substrate to the surface, dictates the sequence of the stack. The foundation is a Glass substrate containing oxygen, silicon, and trace metals. Above this lies a dual-layer Buffer composed of a 50 nm

silicon nitride ( $SiN_x$ ) film and a 300 nm silicon dioxide ( $SiO_2$ ) film. This is followed by a 45 nm Polycrystalline Silicon (Poly-Si) active channel layer. The gate stack incorporates a 130 nm

$SiO_2$  Gate Insulator 1 (GI1), a 300 nm Molybdenum (Mo) Gate Electrode 1 (GE1), a 130 nm

$SiN_x$  Gate Insulator 2 (GI2), and a 300 nm Molybdenum Gate Electrode 2 (GE2). Above the

gate architecture lies a 300/200 nm  $SiO_2/SiN_x$  Interlayer Dielectric (ILD) and a 350 nm

$SiN_x$  Passivation (PV) layer.

Interspersed within these upper dielectrics are complex metallic routing layers, specifically Source/Drain 1 (SD1) and Source/Drain 2 (SD2), each configured as a tri-layer Titanium/Aluminum/Titanium (Ti/Al/Ti) stack with respective thicknesses of 80/600/50 nm and 80/700/50 nm. The macro-structure is planarized by massive organic layers, including

Planarization 1 (PLN1) at 2000 nm, and a combined Planarization 2/3 (PLN2/3) layer at 2700 nm. An Anode layer consisting of a 7/110/10 nm Indium Tin Oxide/Silver/Indium Tin Oxide (ITO/Ag/ITO) stack is deposited over the planarization films, capped finally by organic Bank1 (500 nm) and Bank2 (1000 nm) structures.

| Structural Sequence (Bottom to Top) | Target Material Composition | Design Thickness (nm) | Expected Visual Contrast Profile      |
|-------------------------------------|-----------------------------|-----------------------|---------------------------------------|
| Glass Substrate                     | $O/Si$ and Metals           | Unspecified           | Dark, amorphous base                  |
| Buffer                              | $SiO_2/SiN_x$               | 300 / 50              | Dark, continuous dielectric band      |
| Poly-Si                             | Polycrystalline Silicon     | 45                    | Distinct ultra-thin crystalline band  |
| GI1 / GE1                           | $SiO_2$ , $Mo$              | 130 / 300             | Dark dielectric / Bright high-Z metal |
| GI2 / GE2                           | $SiN_x$ , $Mo$              | 130 / 300             | Dark dielectric / Bright high-Z metal |
| ILD                                 | $SiO_2/SiN_x$               | 300 / 200             | Thick dark dielectric composite       |
| SD1                                 | $Ti/Al/Ti$                  | 80 / 600 / 50         | Highly bright metallic tri-layer      |
| PV                                  | $SiN_x$                     | 350                   | Dark dielectric                       |
| PLN1                                | Organic Material            | 2000                  | Massive, continuous dark region       |
| SD2                                 | $Ti/Al/Ti$                  | 80 / 700 / 50         | Highly bright metallic tri-layer      |

|               |                   |              |                                       |
|---------------|-------------------|--------------|---------------------------------------|
| PLN2/3        | Organic Material  | 2700         | Massive,<br>continuous dark<br>region |
| Anode         | <i>ITO/Ag/ITO</i> | 7 / 110 / 10 | Bright conductive<br>composite        |
| Bank1 / Bank2 | Organic Material  | 500 / 1000   | Massive,<br>continuous dark<br>region |

## Resolving the Ambiguity of Indistinguishable Boundaries

The critical metrological challenge explicitly highlighted in this exemplar is that adjacent inorganic layers (e.g., Buffer, GI1, GI2, ILD, PV) and adjacent organic layers (e.g., PLN1, PLN2, PLN3, Bank1, Bank2) exhibit virtually identical gray-level contrast within the SEM micrograph. Consequently, their physical boundaries cannot be visually distinguished based purely on pixel intensity.

To autonomously label the red rectangle, the MLLM must replicate the human methodology outlined in the supporting text: it must identify the layers by cross-referencing their composition, relative color (contrast), and relative thickness against the provided design table. The MLLM begins by anchoring its analysis on the high-contrast elements. The heavy metals, such as the Molybdenum gate electrodes (GE1, GE2) and the Ti/Al/Ti source/drain interconnects (SD1, SD2), scatter electrons heavily and appear as distinct, bright white bands within the cross-section.<sup>8</sup> The model identifies the lowest bright band as GE1, the intermediate bright bands as GE2 and SD1, and the upper massive bright structure as SD2.

Once these high-contrast anchors are established, the MLLM utilizes spatial reasoning and sequence priors to deduce the location of the indistinguishable dark layers.<sup>6</sup> The model reads the table to understand that the dark region immediately beneath the GE1 anchor must be the GI1 layer, and the ultra-thin band below that must be the Poly-Si. Furthermore, when confronted with the massive dark expanse above SD2, the MLLM refers to the table to deduce that this region contains PLN2/3, the Anode, and Bank1/Bank2. By calculating the ratio of the design thicknesses (e.g., recognizing that PLN2/3 is 2700 nm while Bank1 is only 500 nm), the MLLM can estimate the proportionate spatial location of these conceptually distinct layers even in the absence of a visible boundary, mapping the textual labels to the corresponding coordinate space within the red rectangle.<sup>6</sup> This demonstrates a profound level of cognitive fusion, where the model compensates for visual deficiencies by imposing logical constraints derived from tabular text.

## The Agentic Pipeline for Automated Labeling

Translating this cognitive reasoning into an automated software pipeline requires a sophisticated

agentic framework. In highly scaled manufacturing environments, the time taken to process visual information is critical.<sup>1</sup> The workflow for labeling the layers within the designated red rectangle leverages the concepts of in-context learning, multimodal prompting, and speculative execution.

## Multimodal Instruction Tuning and Prompt Architecture

To ensure the MLLM correctly interprets the semiconductor micrograph, it must be conditioned through physics-aware instruction tuning.<sup>16</sup> Frameworks such as QuPAINT and datasets like QMat-Instruct have been developed to provide MLLMs with multimodal, physics-aware question-answer pairs.<sup>16</sup> This training teaches the model to understand the distinct visual signatures produced by specific material compositions under an electron beam.<sup>13</sup> During inference, the model is tested on its ability to recognize that high-Z (high atomic number) elements appear bright in Z-contrast imaging, while low-Z elements like organic polymers or  $SiO_2$  appear dark.<sup>8</sup>

The human operator initiates the process by supplying the image, drawing the red rectangle bounding box, and providing the supporting data table as a combined prompt.<sup>9</sup> The prompt architecture utilizes few-shot or one-shot learning techniques.<sup>9</sup> By providing a previous example where a human has manually drawn lines connecting textual labels to specific striations, the model understands the structural format required for the output.<sup>6</sup> The red rectangle acts as a powerful spatial prior, acting as an attention mechanism that forces the model's transformer layers to calculate pixel similarities and semantic features exclusively within that bounded region, ignoring extraneous background data.<sup>9</sup>

## Overcoming Agentic Depth and Sequential Latency

Querying an MLLM to perform multi-step reasoning—first parsing the table, then locating the metal anchors, then deducing the organic sequences, and finally projecting textual labels onto the image coordinates—introduces substantial computational overhead. This cascaded loop of perception, reasoning, and tool invocation is referred to as "agentic depth".<sup>15</sup> In a high-throughput failure analysis laboratory, prohibitive latency is unacceptable.<sup>15</sup> To break this sequential bottleneck, system architectures deploy speculative acceleration frameworks, such as SpecEyes.<sup>15</sup> The core innovation here involves utilizing a lightweight, tool-free MLLM to act as a speculative planner.<sup>15</sup> This smaller model rapidly predicts the likely execution trajectory and the spatial location of the major layers. This prediction guides the heavy, full-parameter MLLM, allowing it to bypass exhaustive pixel-by-pixel semantic correlation and focus its heavy computational power solely on resolving the complex, ambiguous boundaries, such as separating the Buffer layer from the Poly-Si.<sup>8</sup> This speculative approach significantly accelerates the generation of the labeled output image.

## High-Precision Segmentation: From CNNs to Vision Foundation Models

Semantic labeling represents only the initial phase of the requested workflow. To execute the future steps of estimating layer thickness and detecting anomalies, the system must transition from merely placing text labels to extracting precise, pixel-perfect mathematical boundaries for every identified layer.<sup>20</sup>

## The Degradation of Legacy Convolutional Architectures

Historically, the semiconductor industry relied heavily on Convolutional Neural Networks (CNNs), such as Faster R-CNN, SegNet, and U-Net, to automate defect classification and semantic segmentation.<sup>11</sup> While these architectures demonstrate proficiency in extracting visual features from large, homogeneous datasets, they suffer from debilitating limitations when applied to the dynamic realities of modern multi-layer film stacks.<sup>11</sup>

CNNs are fundamentally limited by their demand for massive, exhaustively annotated training datasets, often requiring thousands of labeled examples per specific defect class or layer structure.<sup>11</sup> When a manufacturing line transitions from an LTPS display array to an advanced logic node featuring Hafnium Zirconium Oxide (HZO) ferroelectric capacitors, the visual paradigm shifts completely.<sup>24</sup> An HZO stack involves entirely different contrast dynamics, featuring ultra-thin 6 nm HZO layers sandwiched between Titanium Nitride (TiN) and Platinum (Pt) electrodes, often complicated by atomic-scale interfacial oxides like  $WO_x$  or  $TiO_2$ .<sup>24</sup> A CNN trained on LTPS arrays will catastrophically fail on an HZO stack, requiring extensive and costly retraining.<sup>11</sup> Furthermore, legacy CNNs operate with a limited semantic understanding; they classify patterns based on pixel gradients without any cognitive comprehension of the underlying material physics, making them incapable of utilizing tabular supporting data.<sup>11</sup>

## The Implementation of the Segment Anything Model (SAM-2)

To resolve these severe structural limitations, modern agentic workflows discard legacy CNNs in favor of Vision Foundation Models (VFM), most notably the Segment Anything Model (SAM-2) and its derivatives.<sup>10</sup> SAM-2 is engineered for open-vocabulary, zero-shot, and one-shot segmentation, completely eliminating the need for domain-specific retraining.<sup>9</sup>

The integration between the MLLM and SAM-2 operates through a seamless prompt-mask cycle.<sup>9</sup> Once the MLLM has deduced the sequence and approximate location of the layers within the red rectangle, it autonomously generates targeted prompts for SAM-2. These prompts can take the form of textual descriptors (e.g., "Segment the bright metallic band corresponding to SD2") or geometric point prompts (e.g., feeding the exact pixel coordinates representing the center of the GE1 layer).<sup>9</sup> SAM-2 receives these prompts and dynamically generates highly accurate binary segmentation masks that precisely trace the topographical contours of the layers across the entire width of the micrograph.<sup>26</sup>

| Architecture<br>Characteristic | Legacy CNNs (e.g., U-Net,<br>Faster R-CNN) | Foundation Models (e.g.,<br>SAM-2 integrated with |
|--------------------------------|--------------------------------------------|---------------------------------------------------|
|--------------------------------|--------------------------------------------|---------------------------------------------------|

|                               |                                                                    | MLLM)                                                            |
|-------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------|
| Data Dependency               | Prohibitive (Requires thousands of annotated images) <sup>11</sup> | Zero-Shot to Few-Shot capability <sup>26</sup>                   |
| Cross-Domain Adaptability     | Poor (Fails upon introducing new materials/stacks) <sup>11</sup>   | Excellent (Adapts dynamically via prompt injection) <sup>9</sup> |
| Semantic Integration          | None (Operates blindly on pixel intensity) <sup>11</sup>           | High (Guided by MLLM reasoning and tabular data) <sup>9</sup>    |
| Handling Invisible Boundaries | Fails entirely (Cannot segment without clear gradients)            | Approximates using MLLM-derived spatial priors <sup>9</sup>      |

Despite its immense power, applying a general-purpose model like SAM-2 to high-noise electron micrographs requires specialized adaptation. Unmodified SAM models can struggle with the inherently weak semantic cues of subtle semiconductor structures, often resulting in scattered, false-positive binarization.<sup>10</sup> To optimize boundary extraction, specialized frameworks such as IC-DefectNCD deploy Self-Saliency-driven Adaptive Binarization.<sup>10</sup> This technique forces the model to mine intra-image cues and residual localizations, ensuring that the final segmentation masks adhere strictly to the physical material interfaces, effectively filtering out the ambient noise inherent in low-dose TEM or SEM imaging.<sup>10</sup> For layers like the organic PLN sequences where physical boundaries are truly non-existent, the system interpolates the boundary mask based strictly on the proportional mathematical distribution derived from the MLLM's analysis of the design table.<sup>9</sup>

# Metrological Scale Calibration and Optical Character Recognition

The extraction of precise mathematical segmentation masks remains fundamentally useless unless those masks can be mapped to physical dimensions. To execute the future step of estimating layer thickness, the automated system must possess an inherent understanding of the micrograph's spatial scale. In human-driven workflows, the engineer simply reads the scale

bar permanently burned into the footer of the image (e.g., the "2  $\mu m$  " scale bar clearly visible in the LTPS array image) and manually calibrates the measurement software.<sup>5</sup> For an autonomous MLLM agent, this requires the deployment of a highly robust Optical Character Recognition (OCR) and text verification pipeline.<sup>5</sup>



## LLM-Assisted Scale Bar Extraction Frameworks

The automated extraction of scale bars from SEM and TEM images is a complex multi-modal task due to varying fonts, background noise, and heterogeneous image formats.<sup>5</sup> Recent advancements have culminated in the development of multi-modal, LLM-assisted scale bar detection frameworks that operate in discrete, highly optimized phases.<sup>5</sup>

The process initiates with an automated object detection module specifically trained on synthetic datasets generated by Auto-DG algorithms.<sup>5</sup> This module scans the entire micrograph and draws a bounding box exclusively around the physical scale bar line and its adjacent alphanumeric text, achieving exceptional localization metrics, including 100% precision and a mean Average Precision (mAP) of 99.2% at an Intersection over Union (IoU) of 0.5.<sup>5</sup>

Once the scale region is isolated, the system deploys a hybrid OCR engine utilizing advanced algorithms such as DenseNet and Convolutional Recurrent Neural Networks (CRNN) to extract the text.<sup>5</sup> This hybrid approach significantly outperforms standalone legacy OCR engines, delivering an 89% precision rate on scientific images.<sup>5</sup> However, OCR algorithms are notoriously susceptible to misinterpreting single characters, particularly confusing "nm" (nanometers) with

"mm" (millimeters) or misreading the Greek letter " $\mu$ " (microns).<sup>5</sup>

To guarantee absolute metrological accuracy, the extracted text is passed to an LLM reasoning agent. The LLM evaluates the OCR output against the visual context of the image and its general knowledge of semiconductor physics.<sup>5</sup> If the OCR outputs "10 mm" for a high-resolution TEM image showing individual atomic lattice fringes, the LLM recognizes the physical impossibility of the scale and autonomously corrects the error to "10 nm", verifying the result before passing it to the measurement backend.<sup>5</sup> Alternatively, if the original image is provided as a proprietary digital micrograph (DM) file, Python libraries such as SimpliPyTEM can be directed by the agent to bypass OCR entirely and extract the precise pixel-size calibration directly from the embedded metadata.<sup>34</sup>

### Algorithmic Pixel-to-Metric Conversion

With the physical value of the scale bar established and verified (e.g.,  $2\ \mu\text{m}$ ), the underlying Python backend calculates the "Pixels Per Metric" ratio.<sup>35</sup> The system analyzes the isolated scale bar graphic, calculates its length in pixels (e.g., measuring the line as 400 pixels wide), and derives the conversion factor.<sup>35</sup> This mathematical ratio acts as the fundamental multiplier for all subsequent dimensional analyses performed on the segmentation masks.<sup>36</sup>

## Algorithmic Thickness Metrology and Physics-Based Modeling

With the micrograph fully calibrated and the distinct material layers accurately segmented by SAM-2, the agentic workflow proceeds to the rigorous mathematical estimation of layer thickness.<sup>21</sup> It is crucial to note that at the microscopic level, semiconductor interfaces are rarely



perfectly planar.<sup>38</sup> Effects such as interface roughness, diffusion, and intermixing ensure that a single point-to-point distance measurement is statistically invalid.<sup>38</sup>

## Gray-Level Intensity Profiles

To achieve robust, reproducible thickness estimation, the automated system relies on the generation of vertical line scans, also known as gray-level intensity profiles.<sup>21</sup>

The MLLM instructs the Python image processing backend—typically utilizing libraries such as OpenCV or SciPy—to cast a dense array of parallel vertical lines across the entire width of the segmented layer within the red rectangle.<sup>35</sup> As a virtual line traverses downwards through the image, the software records the grayscale pixel intensity at every point. When the line crosses a boundary from one material to another—for example, transitioning from the heavy Molybdenum

GE1 layer to the lighter  $\text{SiO}_2$  GI1 layer in the LTPS array—the electron scattering cross-section abruptly changes, resulting in a distinct shift in the plotted intensity curve.<sup>40</sup>

The precise mathematical boundaries between the layers are not determined by simple thresholding, but rather by calculating the inflection points of the smoothed intensity profile using first and second-order derivatives.<sup>21</sup> The distance between these calculated inflection points constitutes the thickness of the layer at that specific vertical slice. By aggregating hundreds of these parallel profiles, the system calculates a highly accurate mean thickness, alongside critical statistical variance metrics including standard deviation, minimum/maximum thickness, and the root-mean-square roughness of the interface.<sup>38</sup>

## Compensating for Probe Broadening with JMONSEL

While intensity profiles are sufficient for macroscopic structures like the massive 2700 nm organic PLN2/3 layers, estimating the thickness of ultra-thin films—such as the 6 nm Hafnium Zirconium Oxide (HZO) layers or the sub-20 nm logic gate oxides—requires an understanding of electron beam physics.<sup>12</sup>

When a high-energy electron beam penetrates a relatively thick sample, elastic scattering causes the interaction volume of the probe to broaden as it travels deeper in the Z-direction.<sup>47</sup> Consequently, the secondary electron intensity profile at an edge becomes dependent on both the layer's true width and the specific scattering characteristics of the material, causing narrow lines to appear blurred and artificially thick.<sup>46</sup>

To counteract this, the most advanced MLLM metrology pipelines integrate physics-based simulation engines, such as the Java Monte Carlo Simulation of Secondary Electrons (JMONSEL).<sup>46</sup> Instead of relying solely on the raw image data, the system utilizes JMONSEL to generate a theoretical library of predicted electron yield profiles based on the sample's known geometry and composition.<sup>46</sup> The metrology backend models the central peak of the measured profile using a Gaussian function and describes the scattering tails using exponential decay functions.<sup>47</sup> The software iteratively adjusts these shape parameters until the theoretical model perfectly aligns with the empirical measurement.<sup>46</sup> This profound integration of computational physics ensures that thickness estimation achieves true nanometer-scale accuracy, mitigating

the resolution limits imposed by beam broadening and signal-to-noise degradation.<sup>38</sup>

## Autonomous Anomaly and Defect Detection

The ultimate objective of the requested workflow is anomaly detection. In the context of semiconductor failure analysis, anomalies are not limited to catastrophic mechanical failures like delamination; they encompass a vast spectrum of subtle morphological deviations, including interfacial voiding, unexpected crystallographic phase transitions, and improper film stoichiometries.<sup>28</sup>

### Visual Identification of Morphological Defects

When the MLLM surveys the segmented image, it leverages its instruction-tuned knowledge base to identify visual anomalies that deviate from the expected baseline.<sup>10</sup> Within the context of the LTPS array, the model might identify a continuity break in the Ti/Al/Ti SD1 interconnect, suggesting a catastrophic open-circuit defect resulting from aggressive etching. Alternatively, when analyzing a high-k/metal gate stack, the model can detect the presence of unwanted interfacial oxides, such as a localized  $TiO_2$  formation resulting from oxygen scavenging at a  $TiN/HZO$  interface during thermal annealing.<sup>24</sup>

Furthermore, the system can identify Strong Metal-Support Interactions (SMSI), where partially reduced oxide species migrate to the surface of a metal layer, fundamentally altering the expected grayscale contrast and indicating severe thermal degradation.<sup>51</sup> By utilizing open-vocabulary foundation models like RF-DETR or SAM-2, the anomaly detection module is not constrained to a predefined list of known flaws; it can flag any localized morphological aberration or structural asymmetry as a potential defect for further review.<sup>10</sup>

### Agentic Frameworks for Root-Cause Prognostics

Identifying the physical presence of a defect is merely the diagnostic phase; determining the origin of the defect requires sophisticated root-cause analysis.<sup>1</sup> The most profound application of MLLMs in this domain is their orchestration within multi-agent LangGraph pipelines, such as the SemiFA framework, which transforms simple image inspection into holistic factory intelligence.<sup>1</sup>

The SemiFA architecture decomposes the failure analysis process into a sequence of specialized AI agents:

1. **DefectDescriber:** Utilizing DINOv2 vision embeddings and an MLLM like LLaVA-1.6, this agent classifies the visual morphology of the defect identified in the SEM cross-section, generating a structured narrative of the anomaly.<sup>1</sup>
2. **RootCauseAnalyzer:** This agent represents the nexus of multimodal fusion. It takes the visual data and narrative from the DefectDescriber and correlates it with external, non-visual equipment telemetry, such as SECS/GEM data generated by the manufacturing tools.<sup>1</sup> It also queries a Qdrant vector database to retrieve historically similar defects.<sup>1</sup>

3. **SeverityClassifier:** Based on the location and magnitude of the defect (e.g., a massive void in the Anode versus a minor variation in the Buffer layer), this agent calculates a severity score and estimates the potential downstream yield impact.<sup>1</sup>
4. **RecipeAdvisor:** The final agent utilizes the aggregated data to propose highly specific corrective actions, suggesting adjustments to process parameters—such as altering the deposition temperature or reactive gas flow rates in an Atomic Layer Deposition (ALD) chamber—to prevent the recurrence of the defect.<sup>1</sup>

For example, if the algorithmic thickness metrology reveals that the  $SiN_x$  PV layer is 200 nm instead of its target 350 nm, and the DefectDescriber identifies high surface roughness, the RootCauseAnalyzer can cross-reference this data with SECS/GEM logs to identify a pressure anomaly in the plasma-enhanced chemical vapor deposition (PECVD) tool.<sup>1</sup> This capability elevates the MLLM from a simple image processing utility into an autonomous, proactive yield-management system.<sup>52</sup>

## Second and Third-Order Strategic Implications

The successful deployment of MLLM-driven automated labeling, metrology, and anomaly detection introduces several profound second and third-order implications that will fundamentally alter the operational landscape of semiconductor manufacturing.

The most immediate consequence is the complete displacement of the traditional failure analysis bottleneck. Historically, human cognitive latency—the hours required to interpret a micrograph—limited the throughput of the FA laboratory.<sup>1</sup> By automating the extraction of structural data and generating comprehensive FA reports in under a minute<sup>1</sup>, the analytical bottleneck is eliminated. This shifts the operational pressure entirely to data acquisition, strongly incentivizing fabs to invest in high-throughput, fully automated SEM and TEM robotics capable of feeding a continuous stream of cross-sectional images to the MLLM pipeline.<sup>15</sup>

Furthermore, these agentic frameworks catalyze the democratization of advanced materials characterization. Extracting deep insights from high-resolution electron microscopy historically required the expertise of a PhD-level crystallographer.<sup>8</sup> Platforms like MicroscopyGPT and ChemMiner codify this highly specialized knowledge, embedding it directly into the accessible interface of a Large Language Model.<sup>8</sup> This empowers standard process engineers to extract complex phase information, atomic coordinate reconstructions, and root-cause analyses simply by querying the system with an image and a natural language prompt, massively expanding the analytical bandwidth of the organization.<sup>8</sup>

Finally, the transition to physics-based algorithmic metrology, driven by models like JMONSEL, enforces a level of statistical rigor and reproducibility that effectively eliminates the subjectivity inherent in human-driven measurements.<sup>38</sup> As device geometries continue to scale into the angstrom era, this guaranteed metrological precision, coupled with the open-vocabulary adaptability of MLLMs, ensures that manufacturers can maintain stringent process controls, accelerate the yield curve for novel architectures, and secure a dominant position in the increasingly complex landscape of microelectronic fabrication.

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